

MUTHUKRISHNAN P

MERN Stack Developer

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PROFESSIONAL SUMMARY

MERN Stack Developer with 5.5+ years of web development experience and strong hands-on work in React.js, TypeScript, Node.js, Express.js, MongoDB, Mongoose, REST APIs, JWT authentication, dashboard development, form workflows, file handling, payments, testing, and production-ready UI/API integration. Experienced in building full-stack business systems with reusable React components, secure Express APIs, MongoDB data models, role-based access, reports, invoices, and operational dashboards. Also brings broader production experience from Symfony/PHP systems, geospatial platforms, and enterprise workflow applications.

TECHNICAL SKILLS

Frontend: React.js, TypeScript, JavaScript, Vite, HTML5, CSS3, Tailwind CSS, Bootstrap 5, Radix UI, Responsive Design
Backend: Node.js, Express.js, TypeScript, REST APIs, Middleware, Service Layer, Controllers, Route Modules
Database: MongoDB, Mongoose, Schema Design, Data Modeling, Query Optimization, Transactions
Authentication & Security: JWT, bcrypt, Role-Based Access Control, CORS, Rate Limiting, Input Sanitization, XSS Protection
MERN Features: React Context, Protected Routes, API Services, Forms, Dialogs, File Uploads, Reports, Payments, Invoices, Dashboards
Testing & Tooling: Vitest, Jest, Supertest, React Testing Library, pnpm, npm, ESLint, Postman, Git, GitHub, Docker
Integrations: Razorpay, Nodemailer, Multer, PDFMake, ExcelJS, QRCode, CSV Parser

PROFESSIONAL EXPERIENCE

Senior Web Developer – Semansys IT Services Pvt Ltd Feb 2022 – Present
Puducherry, India

Tech Stack: React.js, TypeScript, Node.js, Express.js, MongoDB, Mongoose, REST APIs, JWT, Symfony, PHP, MySQL

- Developed full-stack web applications and API-driven workflows for business management, weather, geospatial, healthcare, inspection, and enterprise domains.
- Built React/TypeScript interfaces with protected routes, layouts, reusable components, forms, dialogs, dashboards, tables, charts, pagination, and API service layers.
- Developed Node.js/Express APIs with route modules, controllers, middleware, MongoDB models, validation, authentication, file handling, and integration services.
- Implemented JWT authentication, role-based access, password hashing, request security, rate limiting, CORS handling, and standardized API responses.
- Designed MongoDB/Mongoose schemas for customers, products, orders, invoices, payments, users, settings, files, and stock history workflows.
- Added frontend and backend test coverage for pages, hooks, contexts, controllers, routes, security, stress, and business workflow behavior.
- Integrated payment, email, file upload, PDF, Excel/CSV import/export, QR code, and reporting workflows into production-style applications.

UI Designer – CADD Center Jul 2021 – Feb 2022
Theni, Tamil Nadu

Tech Stack: HTML, CSS, JavaScript, Responsive Design

- Built responsive web pages and user interface components using HTML, CSS, and JavaScript.

- Converted design ideas and wireframes into functional layouts with attention to usability, spacing, and mobile responsiveness.
- Collaborated with developers to integrate frontend pages with dynamic backend workflows.

SELECTED MERN PROJECTS

UBM – Business Management System

Tech Stack: React.js, TypeScript, Vite, Tailwind CSS, Node.js, Express.js, MongoDB, Mongoose, JWT, Jest, Vitest

- Built a full-stack MERN business management application covering dashboard, customers, products, orders, invoices, payments, users, settings, reports, WhatsApp status, and file management.
- Developed React/Vite frontend with protected routes, authentication context, settings context, app layout, sidebar, header, reusable UI components, dialogs, forms, pagination, charts, invoice views, and image upload flows.
- Created TypeScript API service layers for auth, users, settings, customers, products, orders, payments, invoices, dashboard, and file manager modules.
- Built Express backend with route modules, controllers, middleware, services, and Mongoose models for users, customers, products, categories, orders, invoices, payments, settings, files, stock history, and reset tokens.
- Implemented JWT authentication, bcrypt password security, upload middleware, CORS, rate limiting, Mongo sanitization, XSS protection, environment configuration, and database connection handling.
- Integrated Razorpay payments, Nodemailer email workflows, PDF generation, Excel/CSV import/export, QR code generation, file uploads, and reusable settings services.
- Added Vitest/React Testing Library coverage for frontend pages, contexts, and hooks, plus Jest/Supertest coverage for backend users, products, orders, payments, reports, settings, security, stress, dashboard, and file manager APIs.

Business Dashboard Frontend

Tech Stack: React.js, TypeScript, Vite, Tailwind CSS, Radix UI, React Router, React Query

- Implemented dashboard widgets including stat cards, payment charts, order charts, recent orders, low stock alerts, top products, top customers, storage usage, and WhatsApp status.
- Built reusable dialog components for customers, products, orders, payments, invoices, categories, stock history, bulk image uploads, imports, exports, messages, and confirmations.
- Developed reusable UI primitives for buttons, tables, forms, inputs, dialogs, dropdowns, drawers, tabs, toast notifications, sidebars, date range pickers, pagination, charts, and error boundaries.
- Added protected route handling, logout workflow, mobile-aware layout behavior, sidebar state management, role protection hooks, and consistent API error handling.

Business Management API

Tech Stack: Node.js, Express.js, TypeScript, MongoDB, Mongoose, JWT, Jest, Supertest

- Designed REST API modules for authentication, users, customers, categories, products, orders, invoices, payments, dashboard analytics, settings, and file management.
- Created Mongoose models and controllers for operational workflows including order status, payment tracking, invoice generation, customer history, product images, stock history, and settings.
- Implemented service modules for email, PDF generation, import handling, settings management, upload synchronization, transaction helpers, and Razorpay configuration.
- Added automated API tests and security tests to verify route behavior, authentication, authorization, validation, stress handling, and business-critical endpoints.

ADDITIONAL RELEVANT PROJECTS

- BPMC Migration – Backporch Mail Center: Built TypeScript/React-powered frontend bundles and API documentation views for admin, customer, reports, tools, and API modules; worked on backend/API modernization patterns including JWT authentication, role permissions, database migrations, and service-layer architecture.

- Al Bahar and Earthquakes Platforms: Developed React/TypeScript frontend modules for map-heavy platforms with layers, filters, dashboards, multilingual content, and API-driven data presentation, using SCSS, MapLibre GL, REST APIs, and PHP/Symfony.

CORE MERN STRENGTHS

- React.js and TypeScript application development
- Node.js/Express REST API design
- MongoDB and Mongoose schema modeling
- JWT authentication and secure middleware
- Dashboard, reports, forms, dialogs, tables, and workflow-heavy UI
- API service layers and frontend/backend integration
- File upload, payment, email, PDF, Excel/CSV, and QR integrations
- Vitest, Jest, Supertest, and React Testing Library coverage
- Vite, Tailwind CSS, pnpm, and production build workflows

EDUCATION

Bachelor of Engineering – Mechanical Engineering

Theni Kammavar Sangam College of Technology (TKSCT), Theni | 2016 – 2019

DECLARATION

I hereby certify that the information provided in this resume is true, accurate, and complete to the best of my knowledge. I declare that I have not concealed or misrepresented any facts that are material to this application.

I understand that any inaccuracy or omission in the information provided may be grounds for rejection or termination of my candidature at any stage.

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